SIMATS ENGINEERING
ASSESSMENT TOOL – 2

DATABASE MANAGEMENT SYSTEMS (CSA05)
CO5 – Pseudocode Writing Task (Algorithm Design)

Total Marks: 50 | Each Question: 5 Marks

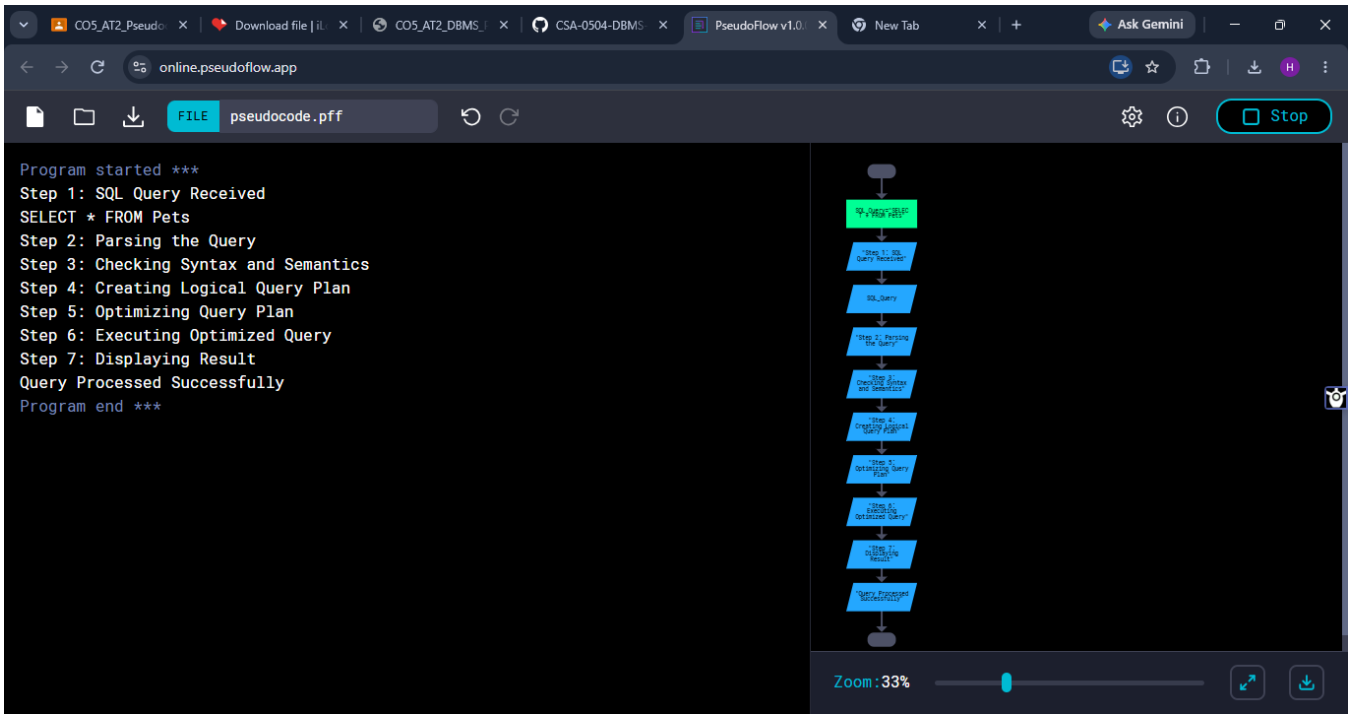
Q1. Write pseudocode for the complete query processing pipeline: from SQL input through parsing, optimization, and execution to result output.

The algorithm accepts an SQL query, checks its syntax and semantics, creates a logical query plan, optimizes it, executes the selected physical plan, and returns the result.

Pseudocode:

```
declare SQL_Query = "SELECT * FROM Pets"


print "Step 1: SQL Query Received"
print SQL_Query


print "Step 2: Parsing the Query"


print "Step 3: Checking Syntax and Semantics"


print "Step 4: Creating Logical Query Plan"


print "Step 5: Optimizing Query Plan"


print "Step 6: Executing Optimized Query"


print "Step 7: Displaying Result"
```

```
print "Query Processed Successfully"
```



OUTPUT

Q2. Design a pseudocode algorithm for cost-based query optimization that estimates and compares I/O costs for different join orderings.

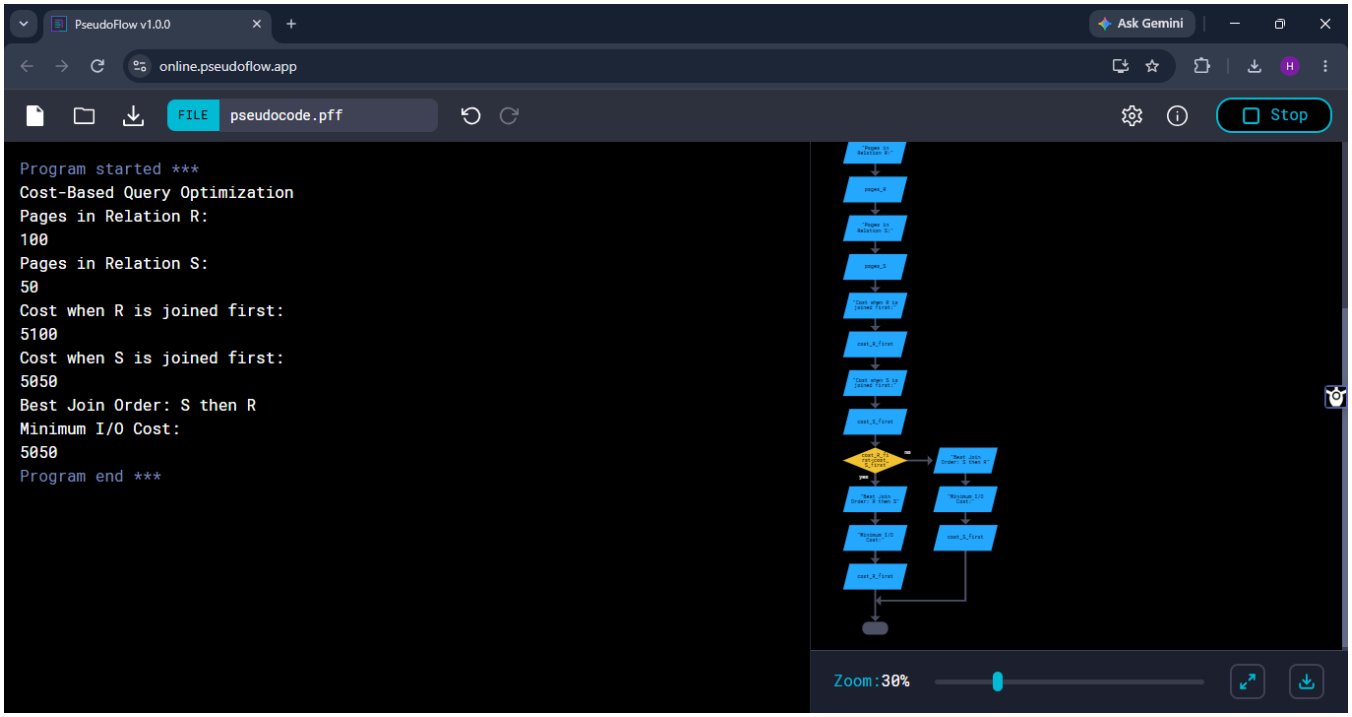
The optimizer evaluates alternative join orders, estimates their I/O costs using relation statistics, and selects the plan with the minimum estimated cost.

Pseudocode:

```
declare pages_R = 100

declare pages_S = 50


declare cost_R_first = pages_R + (pages_R * pages_S)

declare cost_S_first = pages_S + (pages_S * pages_R)


print "Cost-Based Query Optimization"


print "Pages in Relation R:"
```

```
print pages_R

print "Pages in Relation S:"
print pages_S

print "Cost when R is joined first:"
print cost_R_first

print "Cost when S is joined first:"
print cost_S_first

if (cost_R_first < cost_S_first)
    print "Best Join Order: R then S"
    print "Minimum I/O Cost:"
    print cost_R_first
else
    print "Best Join Order: S then R"
    print "Minimum I/O Cost:"
    print cost_S_first
   endif
```

OUTPUT



Conclusion: The optimizer chooses the join ordering having the lowest estimated I/O cost, thereby reducing disk accesses.

Q3. Write pseudocode for the Nested Loop Join algorithm and analyze its time complexity in terms of buffer pages and tuple counts.

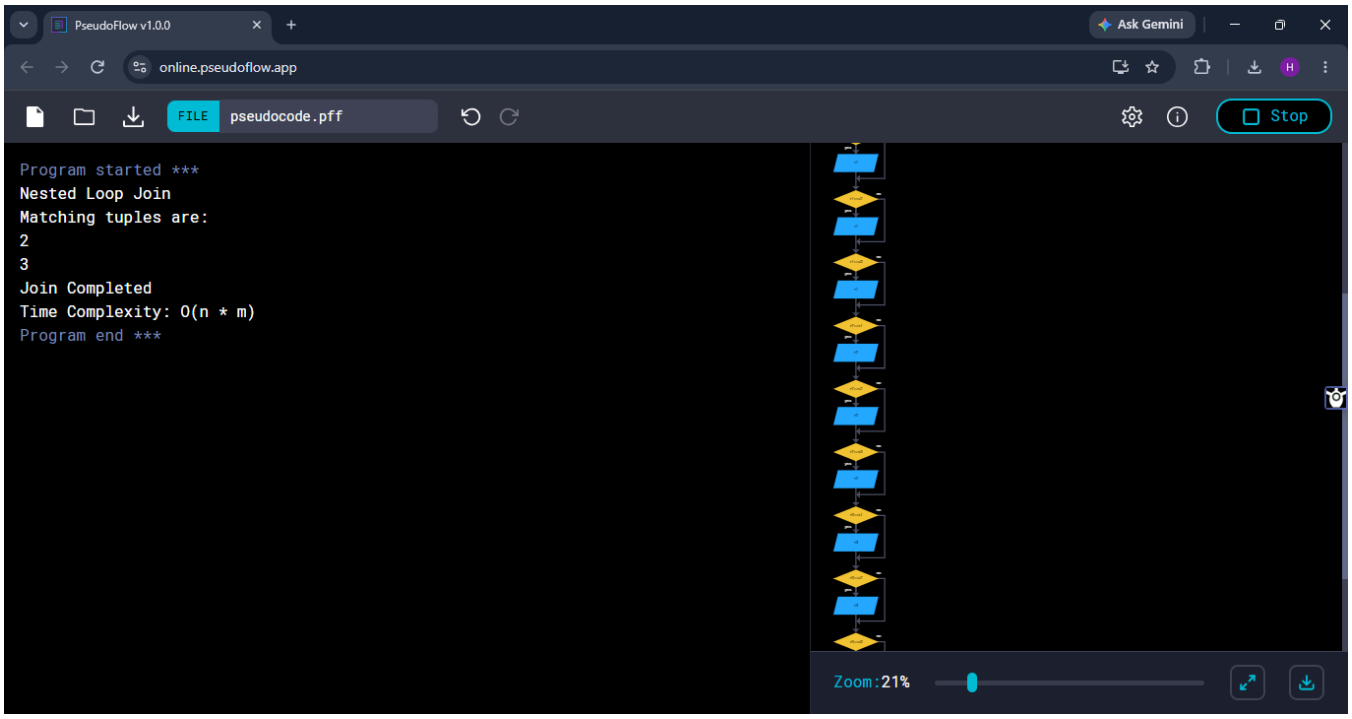
The Nested Loop Join compares every tuple of the outer relation with tuples of the inner relation and outputs matching pairs.

Pseudocode:

```
declare r1 = 1
declare r2 = 2
declare r3 = 3

declare s1 = 2
declare s2 = 3
declare s3 = 4

print "Nested Loop Join"
print "Matching tuples are:"

if (r1 == s1)
    print r1
endif

if (r1 == s2)
    print r1
endif

if (r1 == s3)
    print r1
endif

if (r2 == s1)
    print r2
endif

if (r2 == s2)
    print r2
endif

if (r2 == s3)
    print r2
endif

if (r3 == s1)
    print r3
endif

if (r3 == s2)
    print r3
endif

if (r3 == s3)
```

```
    print r3
endif

print "Join Completed"
   print "Time Complexity: O(n * m)"  END
```

OUTPUT



Complexity Analysis: If R has T_R tuples and S has T_S tuples, tuple comparisons are O(T_R × T_S). If R occupies B_R buffer pages and S occupies B_S pages, a basic page-oriented nested loop can require approximately B_R + B_R × B_S page I/Os (when R is the outer relation).

Q4. Write a pseudocode algorithm to detect deadlocks in a transaction system using a Wait-For Graph (WFG) cycle detection.

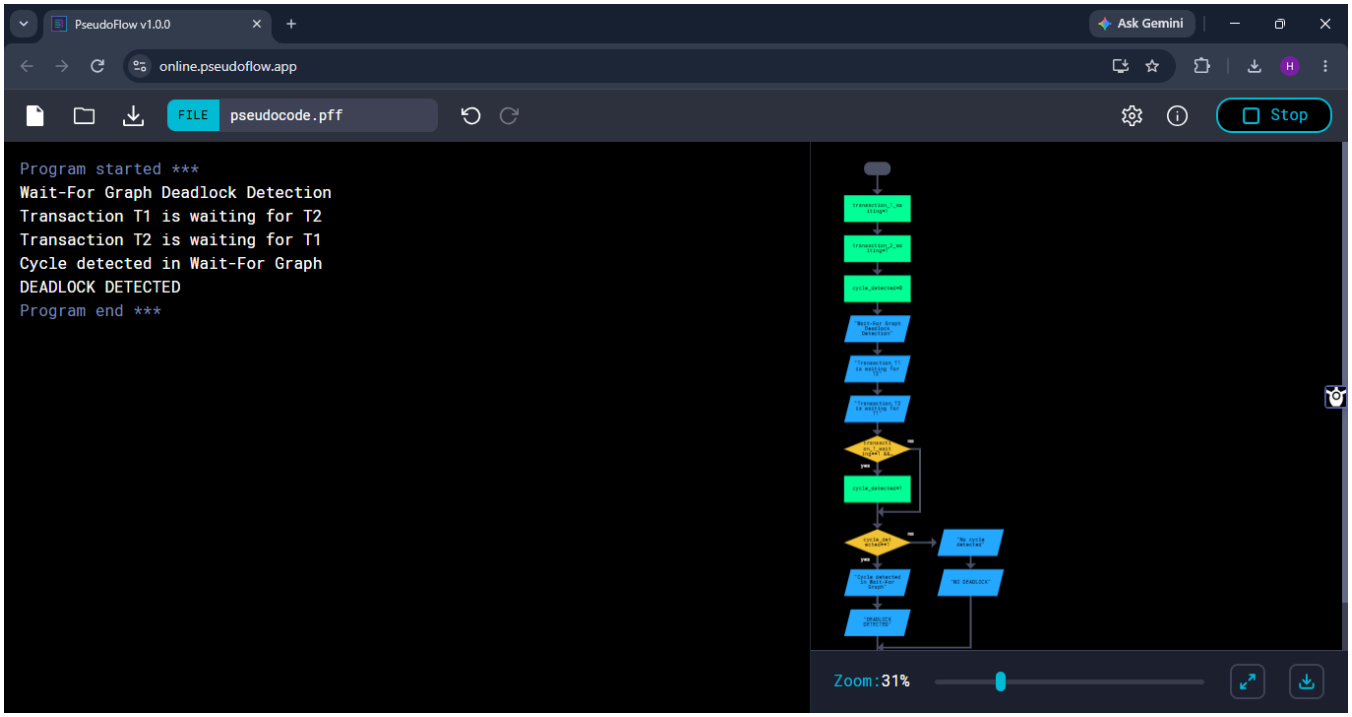
A deadlock exists when the Wait-For Graph contains a cycle. Each node represents a transaction and an edge Ti → Tj means Ti is waiting for Tj.

Pseudocode:

```
   declare transaction_1_waiting = 1

   declare transaction_2_waiting = 1

   declare cycle_detected = 0


   print "Wait-For Graph Deadlock Detection"
```

```
print "Transaction T1 is waiting for T2"
print "Transaction T2 is waiting for T1"

if (transaction_1_waiting == 1 and transaction_2_waiting == 1)
    declare cycle_detected = 1
endif

if (cycle_detected == 1)
    print "Cycle detected in Wait-For Graph"
    print "DEADLOCK DETECTED"
else
    print "No cycle detected"
    print "NO DEADLOCK"
endif
```

OUTPUT



Time Complexity: O(V + E), where V is the number of transactions and E is the number of waiting relationships.

Q5. Design a pseudocode for the Basic Two-Phase Locking protocol including lock request, grant, wait, and release phases.

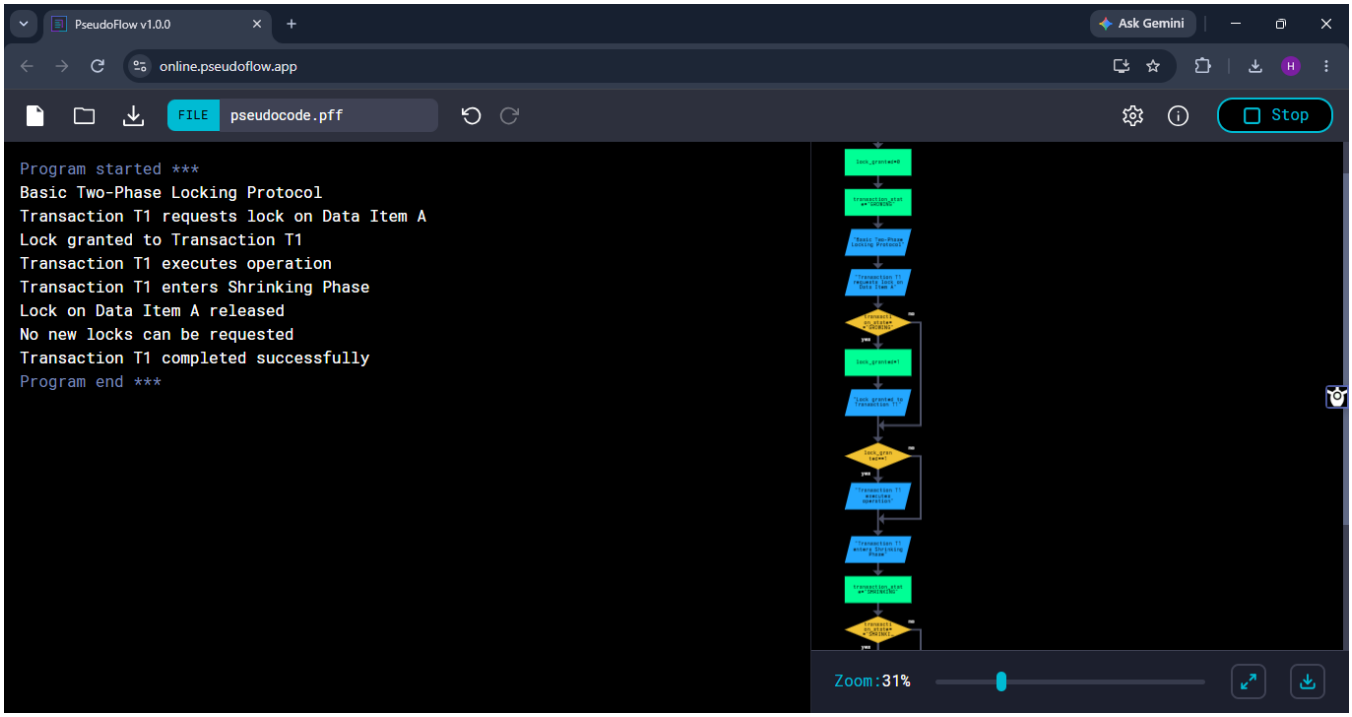
Basic Two-Phase Locking (2PL) has a growing phase in which locks are acquired and a shrinking phase in which locks are released. No new lock is acquired after the transaction starts releasing locks.

Pseudocode:

```
declare lock_granted = 0

declare transaction_state = "GROWING"


print "Basic Two-Phase Locking Protocol"


print "Transaction T1 requests lock on Data Item A"


if (transaction_state == "GROWING")

    declare lock_granted = 1

    print "Lock granted to Transaction T1"

endif


if (lock_granted == 1)

    print "Transaction T1 executes operation"

endif


print "Transaction T1 enters Shrinking Phase"

declare transaction_state = "SHRINKING"


if (transaction_state == "SHRINKING")

    print "Lock on Data Item A released"

    print "No new locks can be requested"

endif
```

```
print "Transaction T1 completed successfully"
```

OUTPUT



Key Rule: Once the transaction enters the shrinking phase, it cannot request any new lock.

Q6. Write pseudocode for timestamp-based concurrency control using Thomas' Write Rule for handling conflicting read/write operations.

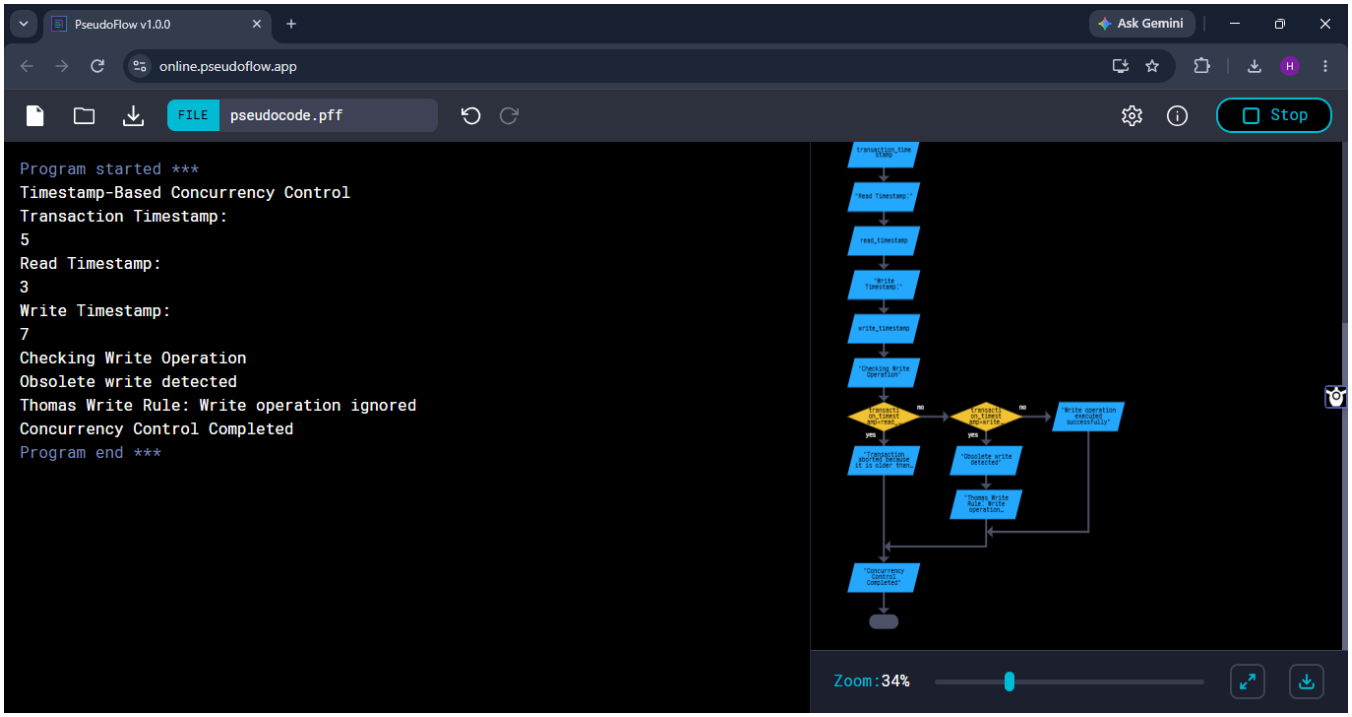
Each transaction receives a unique timestamp. For every data item X, the system maintains Read_TS(X) and Write_TS(X). Thomas' Write Rule allows obsolete writes to be ignored instead of aborting the transaction.

Pseudocode:

```
declare transaction_timestamp = 5

declare read_timestamp = 3

declare write_timestamp = 7


print "Timestamp-Based Concurrency Control"

print "Transaction Timestamp:"

print transaction_timestamp


print "Read Timestamp:"

print read_timestamp


print "Write Timestamp:"

print write_timestamp


print "Checking Write Operation"


if (transaction_timestamp < read_timestamp)

    print "Transaction aborted because it is older than Read Timestamp"

else

    if (transaction_timestamp < write_timestamp)

        print "Obsolete write detected"

        print "Thomas Write Rule: Write operation ignored"

    else

        print "Write operation executed successfully"

    endif

endif
```

```
print "Concurrency Control Completed"
END
```

OUTPUT



Explanation: A write older than an already completed write is obsolete and can be ignored, improving concurrency while preserving timestamp order.

Q7. Write a pseudocode algorithm for the ARIES (Algorithm for Recovery and Isolation Exploiting Semantics) recovery technique with Analysis, Redo, and Undo phases.

ARIES recovery uses write-ahead logging and performs three phases: Analysis to reconstruct transaction state, Redo to repeat history, and Undo to roll back incomplete transactions.

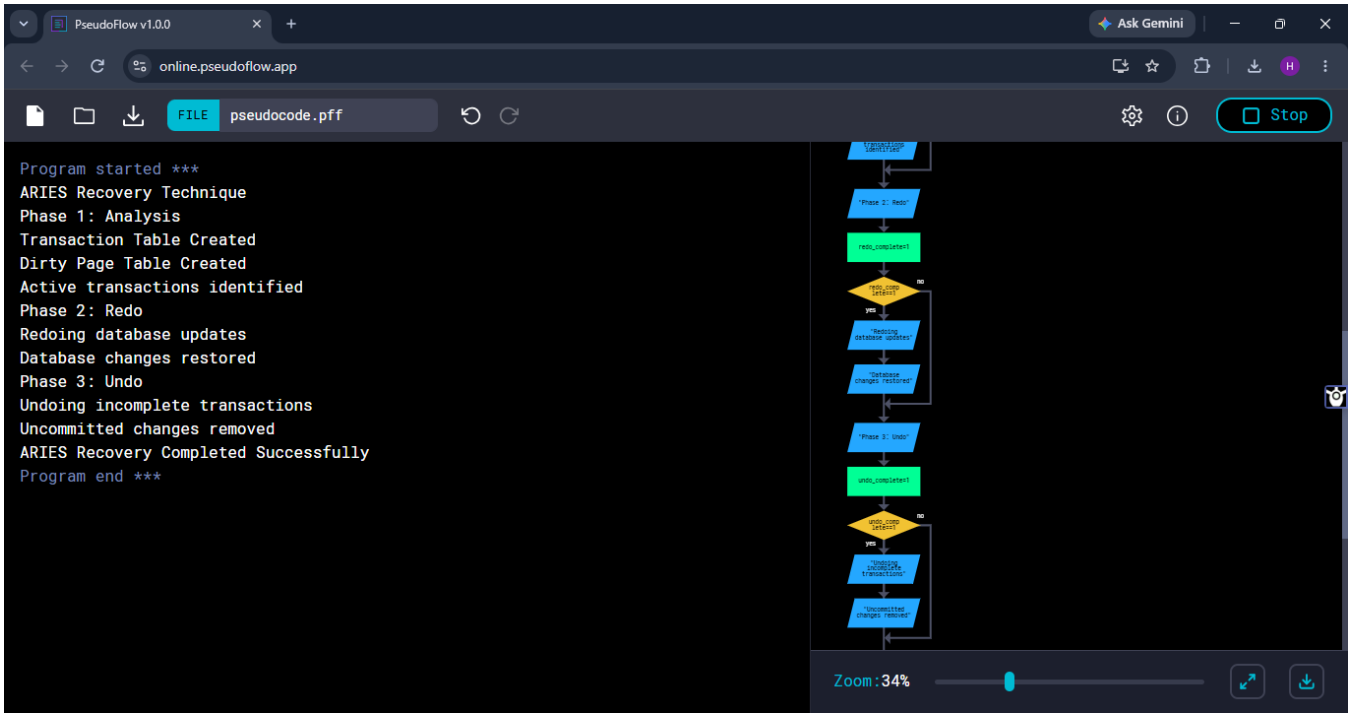
Pseudocode:

```
declare analysis_complete = 0

declare redo_complete = 0

declare undo_complete = 0


print "ARIES Recovery Technique"
```

```
print "Phase 1: Analysis"

declare analysis_complete = 1

if (analysis_complete == 1)
    print "Transaction Table Created"
    print "Dirty Page Table Created"
    print "Active transactions identified"
endif

print "Phase 2: Redo"

declare redo_complete = 1

if (redo_complete == 1)
    print "Redoing database updates"
    print "Database changes restored"
endif

print "Phase 3: Undo"

declare undo_complete = 1

if (undo_complete == 1)
    print "Undoing incomplete transactions"
    print "Uncommitted changes removed"
endif

if (analysis_complete == 1)
    if (redo_complete == 1)
        if (undo_complete == 1)
```

```
            print "ARIES Recovery Completed Successfully"
        endif
    endif
endif
```

OUTPUT



Q8. Design pseudocode for Shadow Paging recovery: maintain current and shadow page tables and roll back on failure.

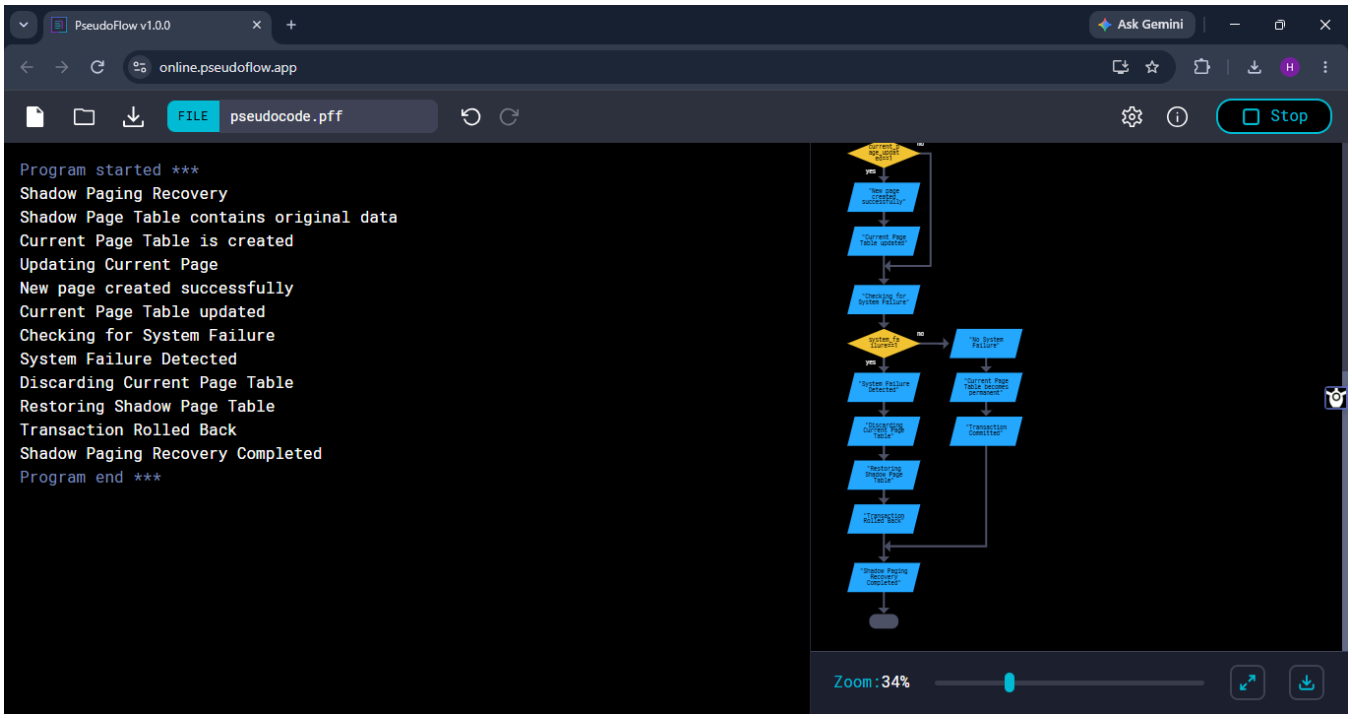
Shadow paging keeps a stable shadow page table and a separate current page table. Updates are made using copy-on-write so the shadow table remains unchanged until commit.

Pseudocode:

```
declare current_page_updated = 0

declare transaction_status = "ACTIVE"

declare system_failure = 1


print "Shadow Paging Recovery"


print "Shadow Page Table contains original data"
```

```
print "Current Page Table is created"

print "Updating Current Page"

declare current_page_updated = 1

if (current_page_updated == 1)
    print "New page created successfully"
    print "Current Page Table updated"
endif

print "Checking for System Failure"

if (system_failure == 1)
    print "System Failure Detected"
    print "Discarding Current Page Table"
    print "Restoring Shadow Page Table"
    print "Transaction Rolled Back"
else
    print "No System Failure"
    print "Current Page Table becomes permanent"
    print "Transaction Committed"
endif
```

OUTPUT



`print "Shadow Paging Recovery Completed"`**Recovery Idea:** After a failure before commit, the unchanged shadow page table directly points to the old consistent database pages.

Q9. Write a pseudocode for Immediate Update recovery using UNDO/REDO log processing after a system crash.

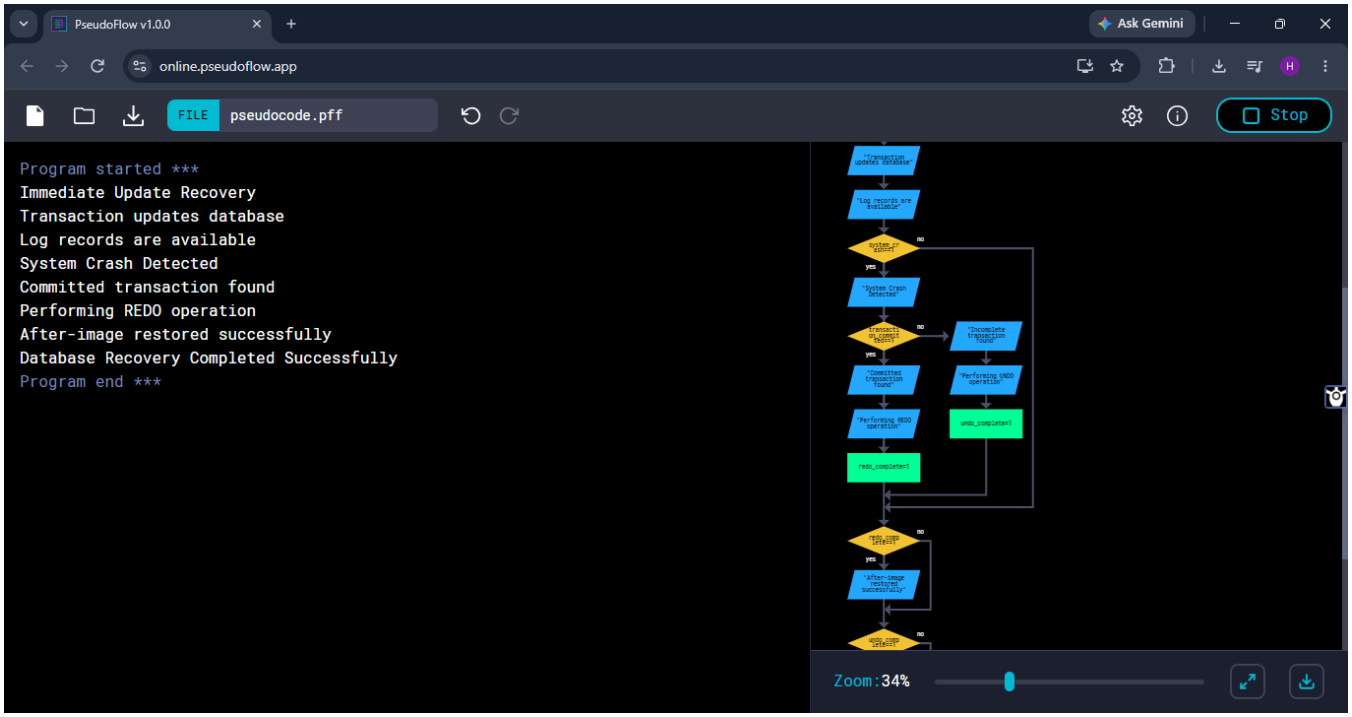
Immediate update permits database pages to be written before commit. Therefore, recovery must REDO committed transactions and UNDO transactions that were incomplete at the time of the crash.

Pseudocode:

```
declare transaction_committed = 1

declare system_crash = 1

declare redo_complete = 0

declare undo_complete = 0


print "Immediate Update Recovery"


print "Transaction updates database"
```

```
print "Log records are available"

if (system_crash == 1)
    print "System Crash Detected"

    if (transaction_committed == 1)
        print "Committed transaction found"
        print "Performing REDO operation"
        declare redo_complete = 1
    else
        print "Incomplete transaction found"
        print "Performing UNDO operation"
        declare undo_complete = 1
    endif
endif

if (redo_complete == 1)
    print "After-image restored successfully"
endif

if (undo_complete == 1)
    print "Before-image restored successfully"
endif

print "Database Recovery Completed Successfully"RETURN "DATABASE RECOVERED"
END
```

OUTPUT



Write-Ahead Logging Rule: The log record containing the required before/after information must be safely written before the corresponding database page is written.

Q10. Write pseudocode for a database connectivity manager that handles connection pooling, query execution, and exception rollback.

The connectivity manager reuses connections from a pool, executes the query inside a transaction, commits successful operations, rolls back on exceptions, and always returns the connection to the pool.

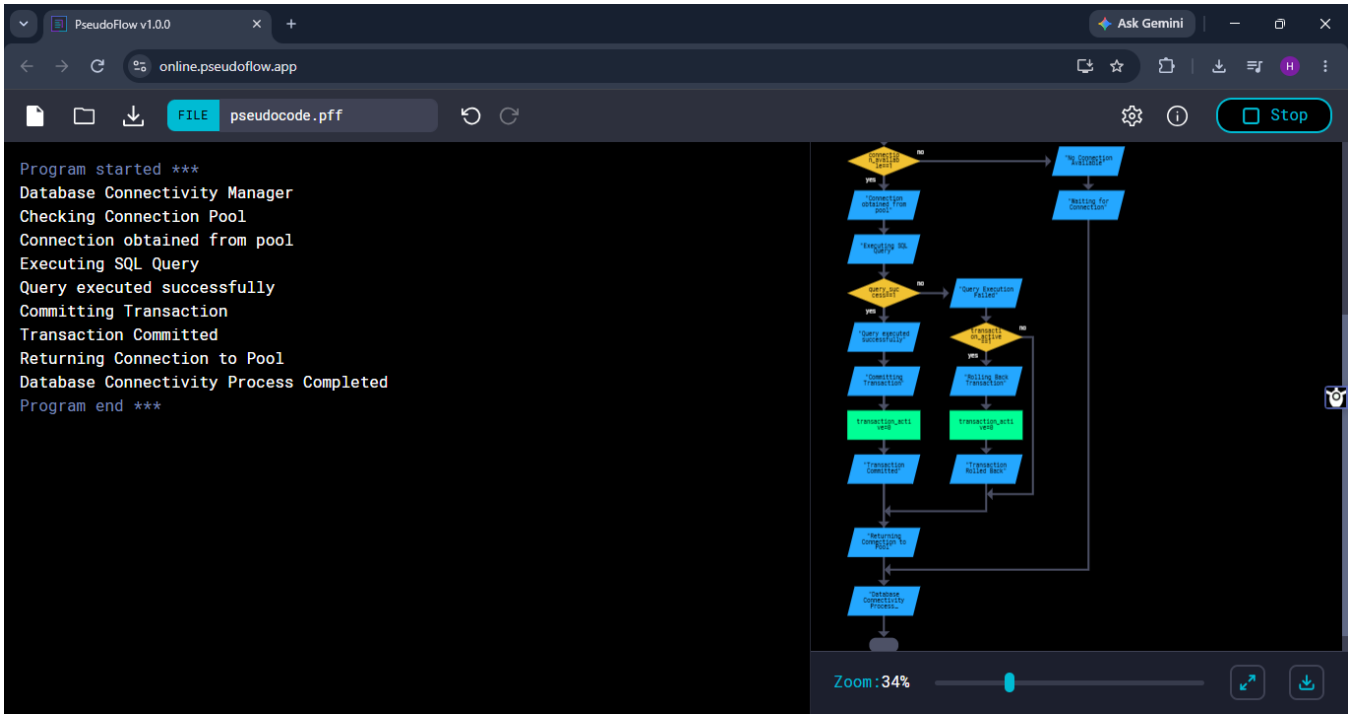
Pseudocode:

```
declare connection_available = 1

declare query_success = 1

declare transaction_active = 1
```

```
print "Database Connectivity Manager"

print "Checking Connection Pool"

if (connection_available == 1)
    print "Connection obtained from pool"

    print "Executing SQL Query"

    if (query_success == 1)
        print "Query executed successfully"
        print "Committing Transaction"
        declare transaction_active = 0
        print "Transaction Committed"
    else
        print "Query Execution Failed"

        if (transaction_active == 1)
            print "Rolling Back Transaction"
            declare transaction_active = 0
            print "Transaction Rolled Back"
        endif
    endif

    print "Returning Connection to Pool"

else
    print "No Connection Available"
    print "Waiting for Connection"
endif
```

```
print "Database Connectivity Process Completed"
```

OUTPUT



Result: The algorithm supports efficient connection reuse, controlled query execution, exception handling, rollback, and safe resource release.

THANK YOU